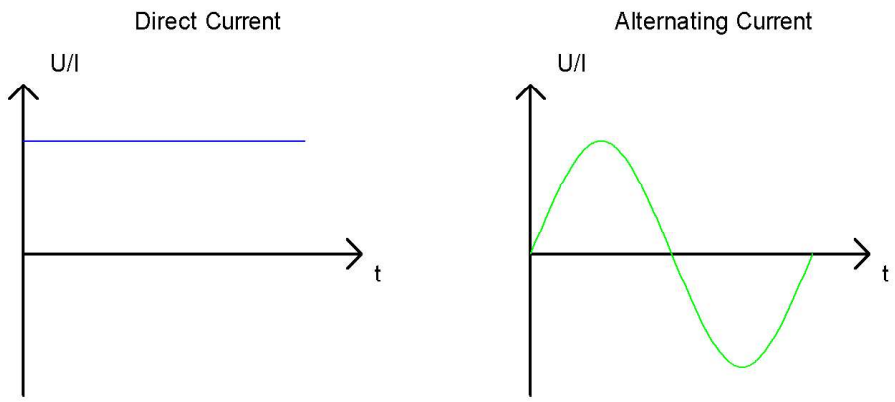
Circuit Knowledge

Power supply

Power supply provides energy for the circuit, and it is divided into DC power and AC power.

Voltage and current of DC power supply remains the same all the time, such as battery, power adapter.

Alternating Current (AC) describes the flow of charge that changes direction periodically. As a result, the voltage level also reverses along with the current. . Its basic form is sinusoidal voltage(current). AC power is suitable for long-distance transmission of electric energy and it is used to supply power to houses.



Generally, electronic circuits use DC. Home appliances have rectifiers to convert AC into DC before they are used.

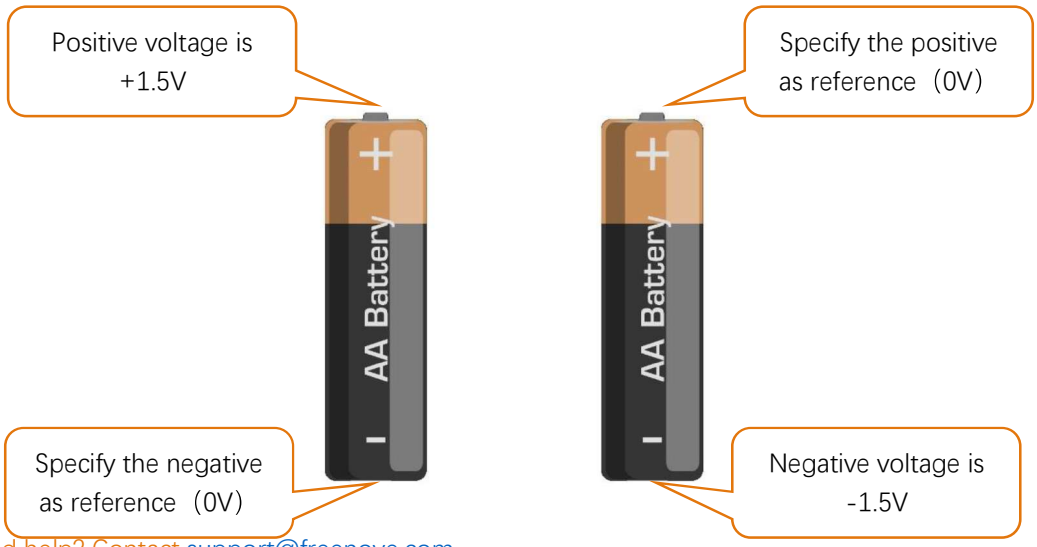
Battery or battery pack can be represented by the following symbols:

The positive and negative poles of the power supply must not be directly connected, otherwise it may scald you and cause damage to the battery.

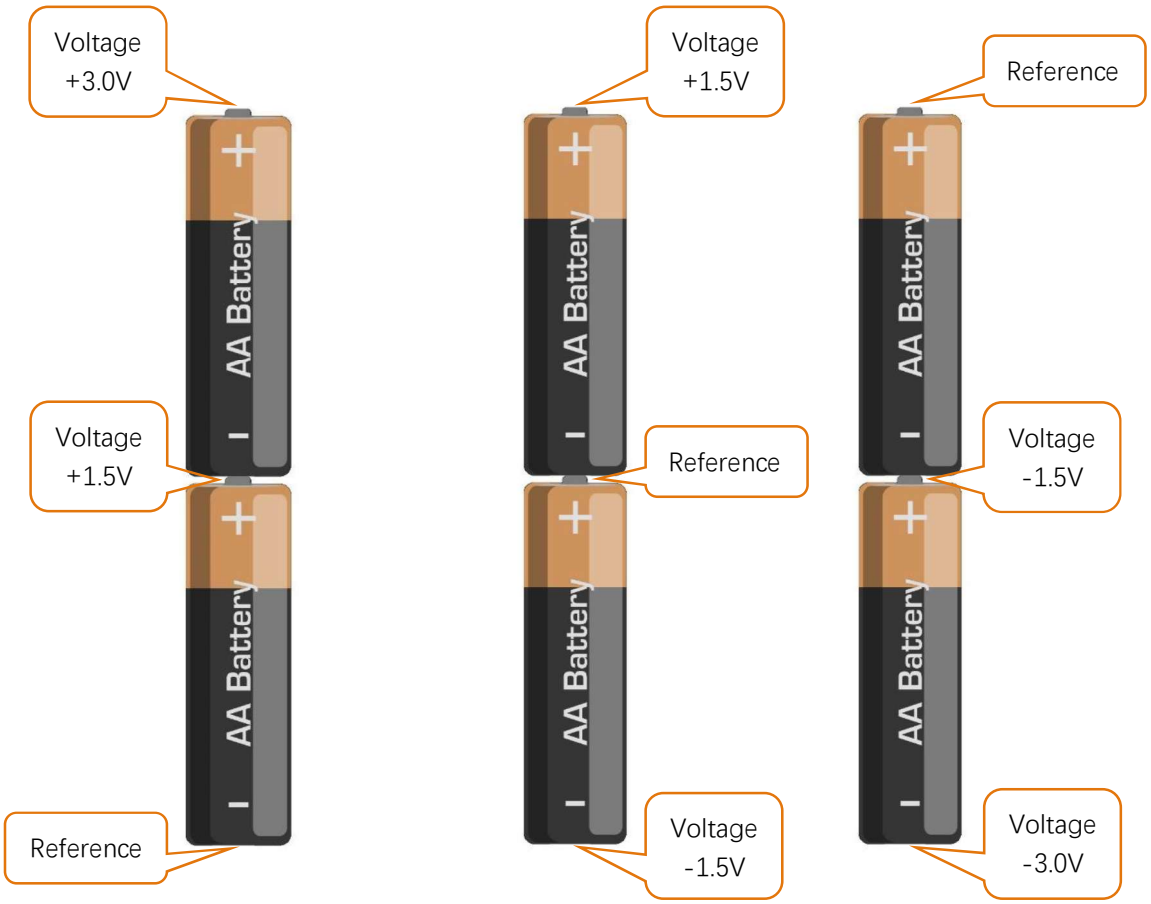
Voltage

The unit of voltage(U) is volt(V). 1kV=1000V, 1V=1000mV, 1mV=1000μv.

Voltage is relative. As to a dry battery marked with "1.5V", it's positive (+) voltage is 1.5V higher than the negative (-) voltage. If you specify the negative as reference（0V）, the positive voltage will be +1.5V.



When two dry batteries are connected in series, the voltage of each point is as follows:



In practical circuits, we usually specify negative as reference voltage (0V), which is called "Ground". The positive is usually called "VCC". The positive and negative poles of power supply is usually represented by the following two symbols:

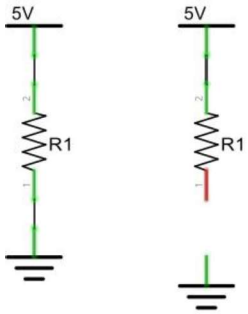
Positive(VCC) Negative(Ground, GND)

Current

The unit of current(I) is ampere(A). 1A=1000mA, 1mA=1000μA.

Closed loop consisting of electronic components is necessary for current to flow.

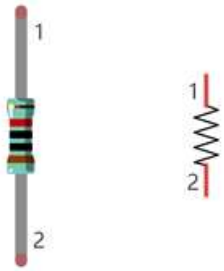
In the figures below: the left one is a loop circuit, so current flows through the circuit. The right one is not a loop circuit, so there is no current.



Resistor

Resistors use Ohms (Ω) as the unit of measurement of their resistance (R). 1MΩ=1000kΩ, 1kΩ=1000Ω.

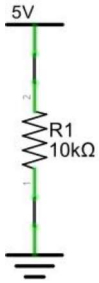
A resistor is a passive electrical component that limits or regulates the flow of current in an electronic circuit. On the left, we see a physical representation of a resistor, and the right is the symbol used to represent the presence of a resistor in a circuit diagram or schematic.



The bands of color on a resistor is a shorthand code used to identify its resistance value. For more details of resistor color codes, please refer to the card in the kit package.

With a fixed voltage, there will be less current output with greater resistance added to the circuit. The relationship between Current, Voltage and Resistance can be expressed by this formula: I=V/R known as Ohm's Law where I = Current, V = Voltage and R = Resistance. Knowing the values of any two of these allows you to solve the value of the third.

In the following diagram, the current through R1 is: I=U/R=5V/10kΩ=0.0005A=0.5mA.



WARNING: Never connect the two poles of a power supply with anything of low resistance value (i.e. a metal object or bare wire) this is a Short and results in high current that may damage the power supply and electronic components.

Component Knowledge

Let us learn about the basic features of components to use them better.

Jumper

Jumper is a kind of wire designed to connect the components together with its two terminals by inserting them onto breadboard or control board.

Jumpers have male end (pin) and female end (slot), so jumpers can be divided into the following 3 types.

Jumper M/M

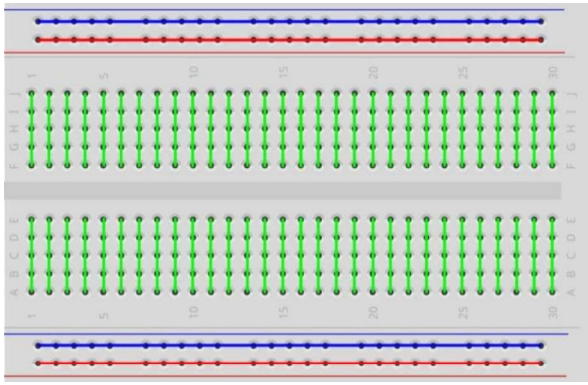
Jumper F/F

Jumper F/M

Breadboard

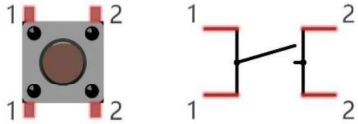
There are many small holes on breadboard to connect Jumpers.

Some small holes are connected inside breadboard. The following figure shows the inner links among those holes.



Push Button Switch

This type of Push Button Switch has 4 pins (2 Pole Switch). Two pins on the left are connected, and both left and right sides are the same per the illustration:

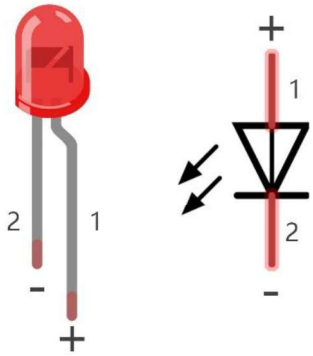


When the button on the switch is pressed, the circuit is completed (your project is Powered ON).

LED

An LED is a type of diode. All diodes only work if current is flowing in the correct direction and have two Poles. An LED will only work (light up) if the longer pin (+) of LED is connected to the positive output from a power source and the shorter pin is connected to the negative (-) negative output also referred to as Ground (GND). This type of component is known as “Polar” (think One-Way Street).

All common 2 lead diodes are the same in this respect. Diodes work only if the voltage of its positive electrode is higher than its negative electrode and there is a narrow range of operating voltage for most all common diodes of 1.9 and 3.4V. If you use much more than 3.3V the LED will be damaged and burn out.



Note: LEDs cannot be directly connected to a power supply, which usually ends in a damaged component. A resistor with a specified resistance value must be connected in series to the LED you plan to use.

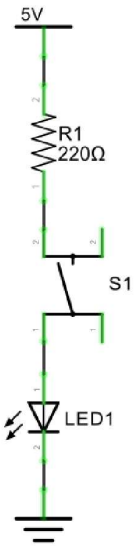
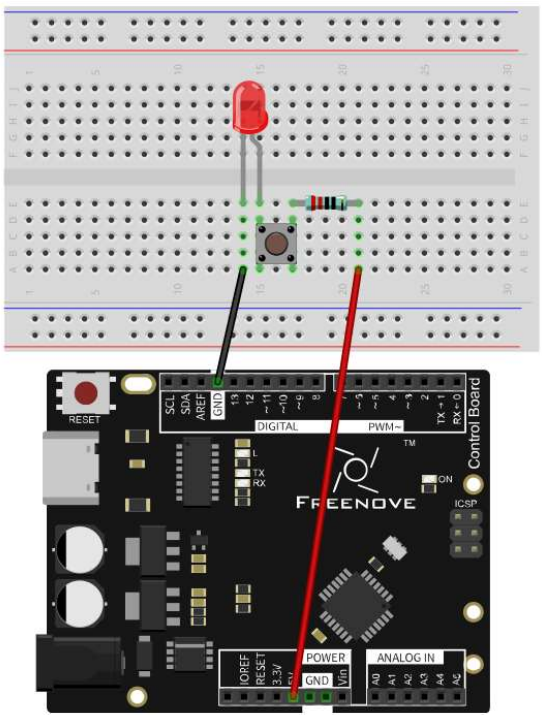
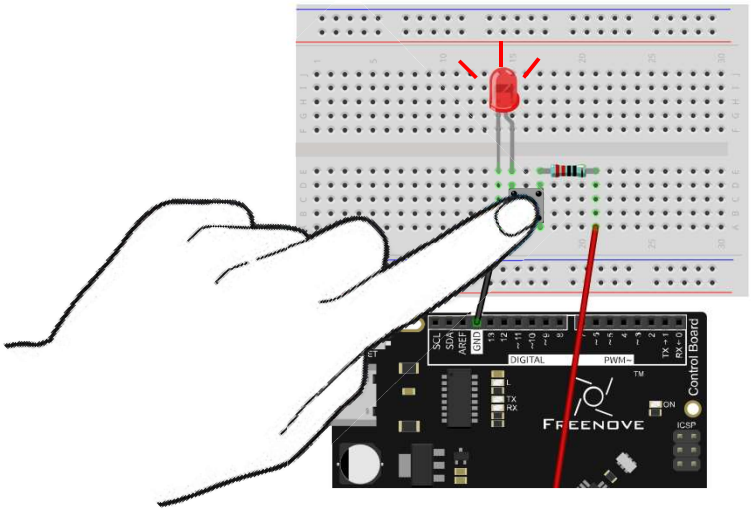
Circuit

In this project, the LED is controlled by a push button switch, and the control board here only plays the role of power supply in the circuit.

Firstly, connect components with jumpers according to "hardware connection". Secondly, check the connection to confirm that there are no mistakes. Finally, connect the control board to computer with USB cable to avoid short circuit caused by contacting the wires.

Note: The control board you received may be black or blue. They are the same in use.

Only the black control board is used to display the hardware connection in this document.

Schematic diagram	Hardware connection
	

LED lights up when you press the push button switch, and it lights off when you release the button.